

SAFE-ALPHA: Anytime-Valid Certification for Adaptive Strategy Search

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Abstract

Adaptive strategy search creates an online multiple-testing problem: proposal identities depend on earlier evaluations, monitoring horizons differ, and the researcher may stop when evidence appears favorable. SAFE-ALPHA separates an arbitrary adaptive proposer from an executable ledger that fixes each strategy, cost rule, bounded score, rank weight, and predictable e-process before its first tested payoff. Joint promotion and proposal-level evidence freezing produce an irreversible set satisfying $\mathbb{E}[\sup_t \text{FDP}(D_t)] \leq q$ under arbitrary serial and cross-strategy dependence. Online e-BH is the testing engine; the contribution is the financial deployment contract and its causal evaluation. In 500 market-shaped sign-null replays at $q = 10\%$, false promotion occurs in 20 runs, while same-bar betting and repeatedly inspected nominal tests fail in every run. At planted annualized Sharpe 1.5 and correlation 0.5, the original persistent rule attains power 0.578, versus 0.479 for a matched terminal rule. A development-frozen finite-campaign prior, daily inspection, and prespecified three-bet mixture reach 0.693 on a new paired holdout, versus 0.572 for the original schedule. A locked U.S. industry replay certifies the market control but none of 76 adaptive symbolic rules; factor residualization removes the control certificate. The evidence concerns the committed bounded score, not raw-return alpha, draw-down, or profitability.

CCS Concepts

• **Applied computing** → **Economics**; • **Mathematics of computing** → **Probability and statistics**; • **Computing methodologies** → *Artificial intelligence*.

Keywords

adaptive strategy search, e-processes, false discovery rate, factor mining, anytime-valid inference, backtest overfitting

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1 Introduction

Consider a research agent that proposes a momentum rule, reads its backtest, changes the lookback, compares the revision with other candidates, and continues until something appears convincing. A held-out period ceases to be held out when its outcome shapes the next proposal, and a conventional test loses its nominal interpretation when it is checked

until rejection. No malicious data mining is required; multiplicity and optional stopping arise from the architecture of the research process itself.

Recent systems use language models to propose, implement, debate, retrieve, and evolve financial signals [3, 6, 9, 11, 12]. Several enforce out-of-sample isolation or high terminal test-statistic hurdles. Those safeguards do not by themselves control error across an adaptively growing stream that is inspected and stopped from the complete research record.

SAFE-ALPHA assigns proposal and promotion to different components. The proposer may use the complete recorded past and may itself be stochastic, language-model based, or adversarially persistent. The ledger fixes the candidate's formula, universe, execution lag, missing-data policy, cost model, payoff transformation, and evidence rule before another payoff is observed. The candidate then begins with unit evidence, and every material revision receives a new identity. A non-generative gate alone may add the strategy to the certified set.

The protocol indexes both proposal time and market time. Candidate j arrives at the adaptive research time τ_j , whereas its evidence evolves over later market dates. A terminal e-BH analysis presumes a fixed decision horizon; preserving every interim peak is also invalid because the running maximum of an e-process need not be an e-value [13]. SAFE-ALPHA instead allocates a summable budget when the proposal arrives, applies the joint rule only to arrived candidates, and freezes one e-valid value at the first joint promotion time. The resulting set is irreversible, which permits immediate deployment, yet its expected worst realized false-discovery proportion remains bounded throughout the research run.

SAFE-ALPHA makes three contributions distinct from the underlying e-BH rule. First, it defines an executable financial-research contract in which proposal identities and times may depend on the complete past, while the formula, execution lag, costs, score, rank weight, and evidence rule are fixed before the first tested payoff. Second, it proves that joint promotion followed by proposal-level freezing yields an irreversible set with $\mathbb{E}[\sup_t \text{FDP}(D_t)] \leq q$ under arbitrary serial and cross-strategy dependence, and it gives pathwise power results for persistence and denser inspection. Third, it evaluates the complete causal protocol with valid matched comparators, controlled timing violations, paired power and delay experiments, and a public replay that separates the prespecified control from adaptively selected rules. Online e-BH, stopped e-values, and worst-time FDR control are established statistical ingredients [4, 15]. The contribution here is the deployment contract that makes those ingredients executable in adaptive financial search.

2 Relation to prior work

Multiple testing has long been central to empirical finance. The Reality Check and SPA address the best model in a fixed candidate family [7, 18]; the Deflated Sharpe Ratio and PBO are retrospective diagnostics for selection and backtest overfitting [1, 2]; and the factor-zoo literature documents why ordinary significance thresholds are too permissive [8]. These methods remain useful, but they target a fixed or retrospectively enumerated search rather than hypotheses whose identities, arrival times, and monitoring horizons depend on the evolving research record.

Our gate builds on e-values [14]. e-BH controls FDR for arbitrarily dependent e-values and includes a sequential-trader example [16]; e-value procedures extend to online hypothesis arrival [4, 20]; and stopped e-BH identifies the global filtration required when evolving evidence streams are monitored jointly [15]. The 2026 online e-closure and donation framework strictly improves baseline online e-BH on streams of fixed e-values [19]. It does not directly supply the jointly stopped ledger used here, in which every proposal carries an evolving global e-process, an individual admission floor, and its own frozen promotion value. We therefore use the baseline whose stopped values can be audited proposal by proposal. Our claim concerns neither e-BH nor worst-time FDR in isolation; it concerns their deployment-safe composition with financial payoffs and the empirical evaluation of the causal failures created by adaptive strategy search.

The need for a proposal-time budget can be seen without any finance-specific distributional assumptions. The following counterexample also explains why running terminal e-BH after the agent decides to stop is not a valid repair.

PROPOSITION 1 (A RANDOM TERMINAL FAMILY CAN HAVE FDR ONE). *Let every hypothesis be null and, independently for $j \geq 1$, set*

$$E_j = \begin{cases} j/q, & \text{with probability } q/j, \\ 0, & \text{otherwise.} \end{cases}$$

Every E_j is an e-value. If proposal generation stops at $J = \inf\{j : E_j = j/q\}$, then $J < \infty$ almost surely and ordinary unweighted e-BH applied to $E_1, \dots, E_J$ rejects with FDR one.

PROOF. Each expectation equals one, but independence and $\sum_j q/j = \infty$ imply that a success eventually occurs almost surely. At that success, the one-rejection e-BH boundary for a family of size J is $J/q = E_J$. Since all hypotheses are null, the resulting nonempty discovery set has false-discovery proportion one. $\square$

This construction does not invalidate terminal e-BH for a genuinely fixed family and horizon. It shows that the family size cannot be chosen by the same evidence later submitted to the test, which is precisely what an open-ended proposer would otherwise be allowed to do.

Table 1: The certification contract. Every row is enforced by the ledger rather than by the proposer.

Stage	Committed or permitted information
Proposal	Formula, universe, lag, costs, score, code hash; global past only
Execution	Point-in-time inputs through $t-1$; realized pay-off at t
Evidence	Post-proposal scores; predictable bet and scale only
Promotion	Joint weighted gate; evidence frozen on admission
Revision	New identifier, proposal time, weight, and unit evidence

3 The SAFE-ALPHA protocol

3.1 One filtration and an immutable ledger

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, P)$ be a discrete-time filtered probability space. The global filtration contains all market observations, agent inputs and outputs, candidate specifications, evidence streams, and gate decisions available by t . Candidate j is proposed at a global stopping time τ_j . Its complete executable specification A_j is $\mathcal{F}_{\tau_j}$ -measurable, and only finitely many candidates have arrived by any finite t . Later proposals may depend on all earlier outcomes. A revision of A_j is a new candidate, not a continuation of j .

For $t > \tau_j$, let $X_{j,t} \in [-1, 1]$ be the one-period net performance score generated by the committed execution and cost rules. The null is

$$H_j : \mathbb{E}_P[X_{j,t} \mid \mathcal{F}_{t-1}] \leq 0 \quad \text{for every } t > \tau_j. \quad (1)$$

This is a conditional no-positive-drift null for the *bounded score*. It is stronger than nonpositive unconditional mean, nonpositive Sharpe, or a zero factor-model intercept. If returns are clipped to construct X , the certificate concerns the clipped score; it must not be described as a certificate for the raw mean return.

Let $\lambda_{j,t} \in [0, 1]$ be $\mathcal{F}_{t-1}$ -measurable and define

$$E_{j,t} = \begin{cases} 1, & t \leq \tau_j, \\ \prod_{s=\tau_j+1}^t (1 + \lambda_{j,s} X_{j,s}), & t > \tau_j. \end{cases} \quad (2)$$

Under H_j , E_j is a nonnegative supermartingale in the *global* filtration. The distinction is essential. A process valid only in its own local history can cease to be valid when another strategy stream reveals information used to stop it [15]. Operationally, a signal using date- t closes trades no earlier than the next implementable price; data vintages, proposal timestamps, code hashes, random seeds, and cost inputs are committed to the ledger before the corresponding outcome is observed; Table 1 records these fields.

The predictability requirement has observable force. Under a symmetric null $X_t \in \{-1, 1\}$, choosing $\lambda_t = \frac{1}{2} \mathbf{1}\{X_t = 1\}$ after seeing X_t gives expected one-step multiplier 1.25,

not at most one. This is the same-bar leakage implemented in our ablation. Likewise, retaining $\sup_{s \leq t} E_{j,s}$ generally spends evidence more than once. By contrast, the first joint-promotion time below is a global stopping time, so its single frozen value remains e-valid by optional sampling.

3.2 Summable budget and irreversible gate

Proposal j receives a nonnegative weight $\gamma_j \in \mathcal{F}_{\tau_j}$ with $\sum_{j \geq 1} \gamma_j \leq 1$ almost surely. Our default,

$$\gamma_j = \{j(j+1)\}^{-1}, \quad (3)$$

supports an unbounded stream. A data-dependent terminal family size cannot safely replace this proposal-time budget. We use the convention $(q\gamma_j k)^{-1} = \infty$ when $\gamma_j = 0$, and impose the ledger-fixed new-admission floor $h = 1/q$. The floor is not needed for FDR control, but it ensures that every promoted strategy has standalone evidence of at least $1/q$; without it, a weak-evidence candidate can become a passenger after a large set of strong discoveries has lowered the joint threshold.

Let D_{t-1} be the finite set promoted before inspection t . The finite adapted eligibility set obeys

$$D_{t-1} \subseteq \mathcal{A}_t \subseteq \{j : \tau_j < t\}, \quad (4)$$

which formally prevents an unproposed candidate, still carrying unit evidence, from entering the gate. Evidence for a promoted candidate is frozen at its promotion time

$$S_j = \inf\{t \geq \tau_j : j \in D_t\}, \quad \tilde{E}_j = E_{j,S_j}.$$

For $j \in \mathcal{A}_t$, form

$$Z_{j,t} = \begin{cases} \tilde{E}_j, & j \in D_{t-1}, \\ E_{j,t}, & j \in \mathcal{A}_t \setminus D_{t-1}. \end{cases}$$

For a proposed set size k , define

$$\mathcal{C}_t(k) = \left\{ j \in \mathcal{A}_t : Z_{j,t} \geq \max\left(h, \frac{1}{q\gamma_j k}\right) \right\}.$$

The maximal feasible size and the new discovery set are

$$k_t^* = \max(\{0\} \cup \{1 \leq k \leq |\mathcal{A}_t| : |\mathcal{C}_t(k)| \geq k\}), \quad (5)$$

$$D_t = \mathcal{C}_t(k_t^*), \quad (6)$$

where $\mathcal{C}_t(0) = \emptyset$ and newly admitted evidence is then frozen. If $d = |D_{t-1}|$, the induction invariant makes $k = d$ feasible. Hence $k_t^* \geq d$, the joint threshold can only fall, and no promotion is withdrawn. If $|\mathcal{C}_t(k_t^*)| > k_t^*$, the same set makes the next integer feasible, contradicting maximality; thus $|D_t| = k_t^*$. Every old member satisfied its threshold when admitted, and every new member satisfies the current one, which yields the invariant

$$j \in D_t \implies \tilde{E}_j \geq \frac{1}{q\gamma_j |D_t|}. \quad (7)$$

as well as $\tilde{E}_j \geq h$ for every member. Freezing at S_j is not the same as retaining $\sup_{s \leq t} E_{j,s}$: only evidence that passes the joint gate at a declared inspection is remembered.

4 Simultaneous error control

Because the proposal itself is random, write N_j for the indicator that the selected candidate is a true null under P . We require N_j to be $\mathcal{F}_{\tau_j}$ -measurable and, equivalently to Eq. (1) on the null event,

$$N_j \mathbf{1}\{\tau_j < t\} \mathbb{E}_P[X_{j,t} \mid \mathcal{F}_{t-1}] \leq 0.$$

This selective formulation permits the agent to choose A_j from the global past. It prohibits selecting A_j with the return intended to test it.

THEOREM 1 (PERSISTENT SAFE-ALPHA CERTIFICATE). *Suppose τ_j is a global stopping time; A_j, N_j, γ_j are $\mathcal{F}_{\tau_j}$ -measurable; finitely many candidates arrive by each finite t ; Eq. (1) holds on $\{N_j = 1\}$; $\lambda_{j,t}$ is predictable; $\sum_j \gamma_j \leq 1$ almost surely; and the finite adapted gate obeys Eq. (4), is nested, and maintains Eq. (7). Without independence assumptions beyond the stated conditional score null,*

$$\mathbb{E}_P \left[\sup_{t \geq 0} \frac{\sum_{j \in D_t} N_j}{|D_t| \vee 1} \right] \leq q. \quad (8)$$

Consequently $\mathbb{E}_P[\text{FDP}(D_T)] \leq q$ for every almost surely finite global stopping time T , including a stopping rule that uses all market data, agent outputs, evidence, and portfolio results.

PROOF. Predictability and the null condition give, on $\{N_j = 1, \tau_j < t\}$,

$$\mathbb{E}_P[E_{j,t} \mid \mathcal{F}_{t-1}] = E_{j,t-1} \{1 + \lambda_{j,t} \mathbb{E}_P[X_{j,t} \mid \mathcal{F}_{t-1}]\} \leq E_{j,t-1}.$$

Thus $N_j E_{j,t}$, viewed after τ_j , is a nonnegative supermartingale starting at N_j : conditioning on $\{\tau_j = r\}$ verifies the claim in the shifted filtration $(\mathcal{F}_{\tau_j+n})_{n \geq 0}$. The promotion time S_j is global because D_t is adapted. For $\bar{E}_j = E_{j,S_j} \mathbf{1}\{S_j < \infty\}$, optional sampling at $S_j \wedge (\tau_j + m)$, followed by conditional Fatou as $m \rightarrow \infty$, yields

$$\mathbb{E}_P[N_j \bar{E}_j \mid \mathcal{F}_{\tau_j}] \leq N_j. \quad (9)$$

Indeed, $N_j E_{j,S_j} \mathbf{1}\{S_j \leq \tau_j + m\}$ is bounded above by the stopped variable and converges to $N_j \bar{E}_j$, which supplies the unbounded-stopping step.

For $D_t \neq \emptyset$, Eq. (7) implies pathwise

$$\text{FDP}(D_t) \leq q \sum_{j \in D_t} \gamma_j N_j \tilde{E}_j \leq q \sum_{j \geq 1} \gamma_j N_j \bar{E}_j.$$

The right-hand side is independent of t . Taking the supremum, applying Tonelli, conditioning on $\mathcal{F}_{\tau_j}$, and using Eq. (9) give

$$\begin{aligned} \mathbb{E}_P[\sup_t \text{FDP}(D_t)] &\leq q \sum_j \mathbb{E}_P[\gamma_j N_j \bar{E}_j] \\ &\leq q \sum_j \mathbb{E}_P[\gamma_j N_j] \\ &\leq q \mathbb{E}_P \left[\sum_j \gamma_j \right] \leq q. \end{aligned}$$

The stopping-time statement follows because the FDP at T is bounded by the displayed pathwise supremum. $\square$

The persistence mechanism has two pathwise power properties.

PROPOSITION 2 (CONTAINMENT UNDER PERSISTENCE AND GRID REFINEMENT). *At an inspection date t , let R_t be the one-shot floor-augmented weighted e-BH set computed from the unfrozen endpoints on the same eligible set, with the same q , weights, and floor. Then $R_t \subseteq D_t$. Moreover, if one SAFE-ALPHA run inspects on a refinement of another run's grid while all evidence paths and ledger fields remain fixed, its discovery set contains the coarser run's set at every common inspection.*

PROOF. For terminal containment, take the union of the persistent run's old set and R_t . Old members pass at the old set size and terminal members pass at $|R_t|$; every member therefore passes at the no-smaller union size because the weighted boundary decreases with set size. The union is feasible for the maximal step-up rule. For grid refinement, induct on the common inspections and apply the same union argument to the refined run's old set and the coarser run's newly feasible set. $\square$

The same argument has a portfolio interpretation. If all promoted strategies receive equal gross sleeve weights, the fraction of gross capital assigned to true nulls equals $\text{FDP}(D_t)$, so its expected worst-time value is at most q . More generally, allocation shares $a_{j,t} \geq 0$ satisfy the same bound if $a_{j,t} = 0$ outside D_t , $\sum_j a_{j,t} \leq 1$, and $a_{j,t} \leq q\gamma_j \bar{E}_j$. A portfolio optimizer that discards or reweights certificates without preserving this inequality receives no capital-level FDR guarantee from the theorem.

The floor supplies a separate candidate-level reading. On the null branch, promotion implies $E_{j,s_j} \geq 1/q$, so conditional Markov and stopped e-validity give $P(S_j < \infty \mid \mathcal{F}_{\tau_j}) \leq q$ on $\{N_j = 1\}$. This marginal statement does not control the probability of any false promotion across the stream; that is why the weighted joint gate remains necessary.

The default telescope is robust to an unbounded stream but need not match a finite research campaign. If a cap M is declared before the first proposal, a valid finite-campaign prior is

$$\gamma_j^{(\eta)} = \frac{1-\eta}{j(j+1)} + \frac{\eta}{M} \mathbf{1}\{j \leq M\}, \quad 0 \leq \eta < 1. \quad (10)$$

Its infinite sum is one, so Theorem 1 applies without change. For $\eta > 0$, its one-discovery boundary improves on the telescope precisely when $j(j+1) > M$ and $j \leq M$; it sacrifices the earliest ranks and makes post-cap boundaries larger by $1/(1-\eta)$. The cap is a ledger field, not the realized proposal count, and unused uniform mass is never reclaimed.

The main proof also quantifies implementation error. If stopped null evidence instead satisfies $\mathbb{E}[N_j \bar{E}_j \mid \mathcal{F}_{\tau_j}] \leq N_j c_j$, where $c_j \geq 1$ is fixed at proposal time and $\sum_j \gamma_j c_j \leq C$, the same calculation yields

$$\mathbb{E}[\sup_t \text{FDP}(D_t)] \leq qC. \quad (11)$$

Thus a bounded evidence inflation has an explicit validity debt, while dividing each stream by a predictable upper bound on its cumulative inflation restores the level- q statement.

5 Empirical design

5.1 Public-data strategy stream

We use daily value-weighted returns for the 30 U.S. industry portfolios and the daily risk-free rate from the Kenneth French Data Library [5]. The report window is January 3, 2007 through May 29, 2026; five prior years initialize proposal scores. Download hashes and the complete processed panel are recorded. Because the current library reconstructs historical series after CRSP updates, this block is an auditable economic illustration, not a point-in-time claim that historical alpha was discovered.

The fixed symbolic grammar contains 89 executable candidates: one equal-weight industry-market sleeve; 48 weekly or monthly cross-industry momentum and reversal spreads; 24 monthly low- or high-volatility spreads; and 16 weekly or monthly time-series momentum or reversal overlays. Signals use 21, 63, 126, or 252 trading-day lookbacks and are lagged before execution. Long-short rules have unit gross exposure. Turnover is half the ℓ_1 change in pretrade-to-target risky weights, charged at 2, 5, or 10 basis points per unit. Risky weights drift after each return; residual capital earns the risk-free rate. Displayed sleeve allocations are restored daily from their drifted weights, and both within-sleeve and allocation turnover are charged under the declared half- ℓ_1 convention.

To isolate the statistical layer from model-specific prompting, the main experiment uses a transparent adaptive proposer. The market sleeve is a predeclared positive control assigned the first budget slot. At experiment start and on the first trading day of each subsequent quarter, the proposer selects the unproposed rule maximizing its trailing five-year annualized Sharpe minus 0.20 times its largest absolute correlation with earlier proposals. This yields 76 adaptive symbolic proposals plus the control (77 total). The replay therefore evaluates the certification layer, not the proposal quality of a particular LLM. A generative proposer can use the same interface only if every executable output is committed before its first scored payoff.

For certification, net return $R_{j,t}^{\text{net}}$ becomes

$$X_{j,t} = \text{clip}\left(\frac{R_{j,t}^{\text{net}}}{4 \max\{\hat{\sigma}_{j,t-1}, 10^{-6}\}}, -1, 1\right),$$

where $\hat{\sigma}_{j,t-1}$ is a lagged 63-day volatility estimate with a lagged expanding fallback. The fixed proposal-time bet is the mean of up to 1,260 available prior bounded scores divided by their second moment, clipped to $[0.01, 0.50]$; first-date histories contain 1,239 scores after scale initialization. On the frozen panel, scale multiples 2, 4, and 8 clip respectively 6.05%, 0.318%, and 0.0099% of finite scores; the corresponding fractions of proposal-time bets at the upper cap

are zero, zero, and 13.0%. The primary multiple of 4 therefore keeps clipping below 0.4% without activating the upper betting cap, while the sensitivity analysis reports the two alternatives. We use Eq. (3), $q = 0.10$, and inspect every fifth trading day. At 5 and 10 basis points we rebuild candidate return, drifted-weight, and turnover paths; the proposal ledger, 2-basis-point scale reference, and bets remain fixed.

5.2 Identification of validity and power

Historical returns do not reveal which nulls are true. We therefore separate calibration from economics.

Cross-sectional sign null. For each of 500 replications, an independent Rademacher sign ε_t multiplies the entire 89-dimensional realized score vector x_t on each date. This leaves $x_t x_t^\top$, and hence all same-date cross-strategy dependence and heteroskedastic magnitudes, unchanged. Because ε_t is independent of the signed past, $\mathbb{E}[\varepsilon_t x_{j,t} \mid \mathcal{F}_{t-1}] = 0$, even when the candidate index was selected adaptively from that past. The experiment does not preserve serial sign predictability and is not presented as a complete market simulator. The proposer reselects adaptively every 63 trading days after assigning the signed market control to the first slot, until the ledger contains 77 proposals. We compare SAFE-ALPHA with (i) an illegal same-bar bettor that wagers only after seeing that the current score is nonnegative, (ii) one-sided 5% t -tests inspected every week after 63 observations, and (iii) one unadjusted one-sided 5% test after a 252-day hold-out for each horizon-eligible proposal (74 of 77). The last comparator diagnoses ignored cross-proposal multiplicity; it is not a multiplicity-adjusted fixed split.

The wild-sign construction preserves realized same-date dependence and volatility magnitudes but removes serial sign structure. We therefore add a fixed synthetic null with standardized Student t_3 innovations, a predictable common GARCH-style scale recursion, cross-candidate correlation 0.5, and clipping to the same score range. In both null designs every procedure sees the identical candidate paths. A nine-point grid from $q = 0.01$ through 0.20 describes sensitivity around $q = 0.10$ on the same paths.

Paired planted alternatives. Each replication has 80 bounded Gaussian-score candidates, of which 16 have positive drift, 504 burn-in days, and a common 1,260-day proposal/evaluation period. The first proposal has the largest burn-in sample mean; every later proposal is the remaining candidate with the largest trailing annualized Sharpe, with no diversity penalty. Up to 50 candidates are selected at 15-day intervals, so strict post-proposal monitoring ranges from 1,259 days for the earliest proposal to 524 for the latest. The reported planted Sharpe is the nominal pre-clipping mean-to-volatility ratio. It ranges from 0.50 to 3.00 in increments of 0.25, at equicorrelations zero and 0.5, with 250 replications in every cell. SAFE-ALPHA and the matched terminal weighted e-BH comparator receive the same scores, proposal order, proposal-time bets, spending weights, and raw e-process paths; only the decision schedule differs. The terminal rule reads the untouched endpoints once, whereas

SAFE-ALPHA inspects every five days and freezes only promoted copies. Ordinary terminal BH, uncorrected 5% tests, and the five largest post-proposal sample means remain diagnostic comparators. We report realized FDP, end-to-end power over all 16 planted alternatives, proposal recall, conditional gate power, and delay from proposal to decision.

Prospective power extension. No market outcome entered power-design selection. On two recorded 300-path development banks, we chose Eq. (10) with $M = 50$, $\eta = .75$, daily inspection, and the equal arithmetic mixture of three e-processes whose proposal-time bets are $\min\{.5, m\hat{\lambda}_j\}$ for $m \in \{1, 2, 4\}$. Such fixed mixtures are established betting constructions, not a novelty claim [10, 17]. We froze this design and then generated six new 250-path cells at Sharpes 1, 1.5, and 2 and correlations zero and .5. The original public replay remains on the telescope, the fixed plug-in bet, and five-day inspection. A valid conservative comparator promotes proposal j only when $E_{j,t} \geq (q\gamma_j)^{-1}$; conditional Ville and $\sum_j \gamma_j \leq 1$ give FWER at most q . Every comparator receives the same score paths, proposal order, weights, betting components, and inspection grid permitted by the comparison.

Economic and selection diagnostics. We report an ungated top-five policy reselected quarterly from the trailing five years; a fixed split trained from 2002 through 2010, validated from 2011 through 2014, and tested from 2015 through 2026; and a deliberately optimistic full-sample top five. For the certified sleeve we estimate post-certification CAPM and Fama-French five-factor plus momentum intercepts by OLS with heteroskedasticity and autocorrelation consistent standard errors using five lags. A second gate predictably removes those six factor returns with rolling betas estimated only from the previous 1,260 dates and at least 252 observations. Across the 74 symbolic proposals with a complete one-year future window, we compare the five-year selection Sharpe with the next-year Sharpe and bootstrap the median change. Finally, we compute White's Reality Check, Hansen's SPA with consistent recentering, 1,999 stationary-bootstrap draws, and mean block length 20, the Deflated Sharpe Ratio, and ten-block CSCV/PBO. The Reality Check retains its circular-block convention, so its p -values are not numerically interchangeable with SPA. These fixed-search diagnostics test economic interpretations that the sequential score certificate itself does not imply.

6 Results

6.1 Calibration, persistence, and power

The original procedure makes at least one false promotion in 20 of 500 same-date sign-null runs, probability 0.040 with 95% Wilson interval [0.026, 0.061]. Across nine levels, the curve rises from 0.006 at $q = .01$ to 0.072 at $q = .20$ and remains below the diagonal. Under the final campaign and betting design, the corresponding $q = .10$ probability is 0.042, interval [0.028, 0.063], versus 0.040 for proposal-wise FWER. The heavy-tailed volatility null gives 0.034 for the original gate and 0.020 for the final design. By contrast, same-bar

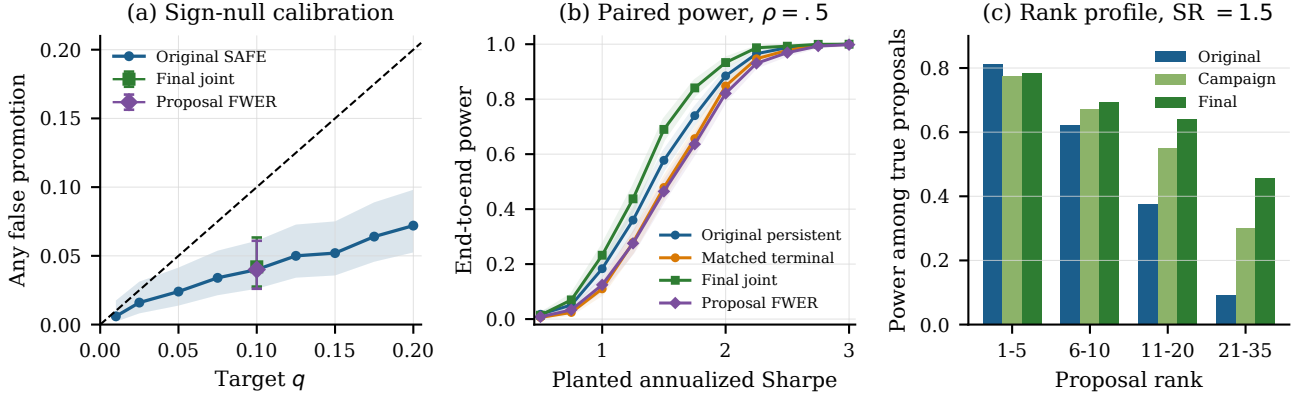


Figure 1: Calibration and paired power. Panel (a) uses 500 sign-null paths; the curve is the original protocol and the two $q = .10$ points use the final design. Panel (b) uses 250 paired paths per effect at $\rho = .5$; shaded regions are seed-level 95% intervals. Panel (c) uses the newly seeded central holdout and counts only true proposals within each rank bin.

betting and repeatedly inspected 5% tests promote in every sign-null run; separate one-year tests across 74 proposals promote in 94.4%. These gray-box ablations identify timing and multiplicity failures, whereas the terminal e-BH and proposal-wise FWER procedures are valid comparators.

At annualized Sharpe 1.5 and $\rho = .5$, the original persistent rule has power 0.578, versus 0.479 for matched terminal weighted e-BH. The paired gain is 0.099, with interval $[0.084, 0.113]$, and persistence never loses on any of the 250 matched paths, as Proposition 2 predicts. Mean decision delay among detected alternatives is 684 trading days for persistence and 1,183 for terminal analysis; the detected subsets differ, so this delay contrast is descriptive rather than a paired causal estimate.

The newly seeded holdout confirms the prospective power extension. At the same central effect, the final design reaches 0.693, compared with 0.645 for campaign weights and daily inspection alone and 0.572 for the original schedule. Its paired gains are 0.048, interval $[0.034, 0.062]$, over the campaign-only design and 0.121, interval $[0.104, 0.137]$, over the original. Against proposal-wise e-alpha-spending on a dense paired grid, joint power is 0.690 versus 0.465, a gain of 0.226 with interval $[0.208, 0.243]$; the joint gate wins on 225 paths and ties on 25. The gain is rank-specific. Relative to the original, final power changes from 0.812 to 0.785 at ranks 1–5, but rises from 0.623 to 0.695 at ranks 6–10, from 0.375 to 0.641 at ranks 11–20, and from 0.091 to 0.457 at ranks 21–35. This is the intended finite-campaign tradeoff, not uniform dominance.

6.2 Locked public replay and selection decay

The rank-one market sleeve is a prespecified pipeline control rather than an adaptively discovered alpha. It crossed its boundary on January 19, 2018 at frozen evidence 20.087,

showing that the proposal, evidence, promotion, and next-bar deployment path can operate on financial data. The substantive discovery result is negative: none of 76 adaptively proposed symbolic rules was certified at $q = 0.10$, and predictable six-factor residualization produced no certificate at all. We therefore read the replay as evidence about selection control in adaptive strategy search, not as evidence that SAFE-ALPHA discovers factor-adjusted alpha. The market control still crossed at 5 and 10 basis points, with the 10-basis-point crossing delayed by one week.

The ledger entry makes the joint rule concrete. At the January 19 inspection, the market control had $\gamma_1 = 1/2$, so a one-member discovery set required

$$E_{1,t} \geq \max\{1/q, (q\gamma_1)^{-1}\} = \max\{10, 20\} = 20.$$

Its current value was 20.087, while no other eligible candidate supported a larger feasible set. The gate froze that value, recorded the code and cost specification used to obtain it, and began portfolio deployment on the next tradable date rather than earning the triggering return twice. Figure 2 records the corresponding weighted paths.

Deployment begins one trading day after promotion. From January 22, 2018 through May 29, 2026, the market control earns 9.96% annualized net excess return at 19.77% volatility, Sharpe 0.504, and maximum drawdown -38.70% . These are properties of a broad market sleeve, not evidence of mined alpha; the large drawdown also shows why a score certificate is not a risk guarantee.

The factor evidence in Table 2 clarifies the certificate’s interpretation. Over 2,100 post-certification dates, the market sleeve has a CAPM beta of 0.941; the six-factor regression explains 95.9% of its variation and estimates a market beta of 0.948. Both intercepts are negative and statistically insignificant. When candidate returns are predictably residualized with betas fitted only through the preceding date, the gate

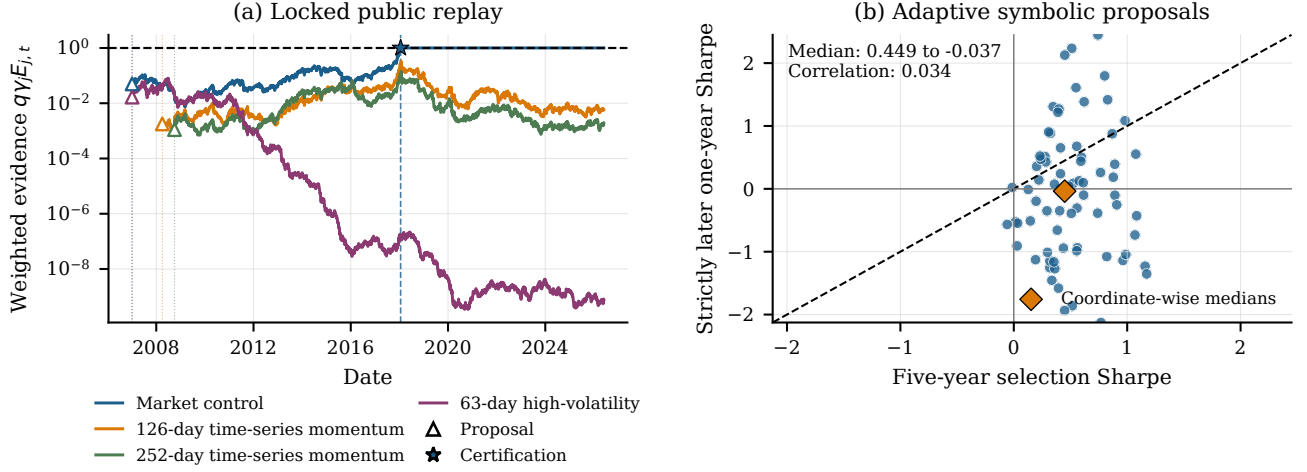


Figure 2: Locked public replay and selection decay. Panel (a) initializes each path at unit evidence on its proposal date and updates it strictly later; the market path is frozen at its crossing. The three symbolic paths are selected for display by their recorded maxima, which never enter the gate. Panel (b) uses the 74 symbolic proposals with a complete 252-day future window; the diamond marks the coordinate-wise medians.

Table 2: Discovery and fixed-family diagnostics. “All” includes the prespecified market control. Reality Check and SPA test different objects from the sequential conditional-score null.

Diagnostic	Adaptive	All
Primary certificates	0 of 76	1 of 77
Factor-residual certificates	0 of 76	0 of 77
Median selected / future SR	.449 / -.037	—
Reality Check p	.190	.022
SPA p	.146	.157

certifies no strategy. The sole primary certificate is therefore a broad market positive control under the committed bounded score, not evidence of factor-adjusted alpha.

The proposal ledger displays the selection effect that a terminal winner table would conceal. Among 74 symbolic proposals with a complete 252-day future window, median annualized Sharpe falls by 0.512, 70.3% perform worse after proposal, only 48.6% remain positive, and the correlation between selection and future Sharpe is 0.034. An independent proposal-resampling 95% percentile interval is $[0.398, 0.754]$, but it is descriptive because the proposals overlap. White’s Reality Check rejects on the full 89-rule panel, but not after removing the market control or restricting attention to adaptive proposals; consistently recentered SPA does not reject in either the full or adaptive family. Although these procedures test different nulls, they point in the same descriptive direction once the prespecified market sleeve is distinguished from the mined symbolic rules.

One-at-a-time design changes reveal where the negative result is stable and where the estimand changes. At $q = 0.05$

no candidate is certified; at $q = 0.20$ the market control and one monthly time-series momentum rule cross. Daily inspection preserves the primary market certificate, inspection every 21 days misses its transient valid crossing, and moving the market control from budget rank one to ranks 5, 10, or 20 produces no certification. Scaling returns by two lagged volatilities yields three promotions, while scaling by eight yields none. These are not interchangeable tuning choices after the fact: they define different ledger-fixed scores, budgets, and inspection opportunities, which is why the ledger records them explicitly.

7 Audit and replication record

The released strategy catalog and proposal, evidence, and certification ledgers identify every empirical decision, while tests enforce strict post-proposal updating and next-bar deployment. SHA-256 digests cover inputs, code, and reported outputs. Replication-level records retain the seed map for the 5,500 paired evaluation paths together with null, sensitivity, and portfolio accounting results.

8 Limitations and scope

The theorem applies to Eq. (1), not to a weaker unconditional alpha null. Our bounded transformation changes the estimand, and a production system seeking a raw-return claim should instead use an e-process justified by explicit tail or conditional-moment assumptions. FDR controls membership of the certified set; it does not guarantee profitability, diversification, market capacity, CVaR, or low drawdown. A downstream optimizer must obey the allocation constraint above if capital-weighted false exposure is the target.

The deterministic proposer is an identification choice: the study isolates the gate rather than benchmarking language models, so it does not establish how proposal quality changes under a particular LLM. The public industry series are nontradable research portfolios and not archival point-in-time vintages; costs omit impact, borrowing, and capacity. Planted alternatives impose persistent constant drift in a stationary Gaussian-score design, not alpha decay, regime change, or crowding. Finally, every summable rank prior trades power across proposal positions. The finite-campaign schedule improves middle and late ranks within its declared cap but weakens the earliest ranks and the post-cap stream. Weak persistent effects can still require long monitoring histories.

9 Conclusion

SAFE-ALPHA reserves promotion for a statistical gate that reads only post-proposal payoffs and freezes evidence at a joint crossing. Under the stated conditional null, the persistent set controls worst-time expected FDP despite dependence and adaptive stopping, while a declared campaign prior, daily inspection, and a fixed betting mixture raise central holdout power by 12.1 percentage points relative to the original schedule. The public replay reaches a deliberately restrained conclusion: the prespecified market control crosses, none of 76 adaptive symbolic rules does, and factor residualization removes the control certificate. The practical value is therefore not a promise of profitable alpha, but a defensible separation between attractive research output and evidence strong enough to justify deployment.

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